1. Digital Sovereignty Starts on Your Desktop

Your computer is the place where your letters, photos, accounts and ideas live. *Digital sovereignty* means that you decide what happens there: which programs run, where your data is kept, who can read it and when the system changes. This chapter explains what that means in practice and what amadeOS does about it.

Who decides on your computer?

On many computers today, the answer is: not you alone. A typical desktop

- asks you to sign in with an online account before you can use your own machine,
- sends usage data ("telemetry") to its maker, often switched on by default,
- stores documents in a cloud by default, on servers in another country and under another law,
- changes its look and its rules with every large update, whether you want it or not,
- shows advertising or "suggestions" inside the system itself.

None of this is a law of nature. It is a choice made by the companies that build those systems, and it binds you to them: your files, your habits and your skills depend on one vendor. Moving away becomes expensive. This is what *lock-in* means.

What amadeOS does differently

amadeOS is built so that the answer is: **you**.

Your computer works without an account. You log in with a user name and password that are stored on the computer itself. No online account is needed to start, use or update amadeOS.

Your data stays on your disk. Documents, pictures and music are kept in the usual folders of your home directory. The applications save in open formats on your disk: the Calendar keeps your appointments in a plain iCalendar file, the Text Editor writes plain text in the encoding it found, WritePad is meant to save OpenDocument files. Nothing is copied to a cloud unless you put it there yourself.

No telemetry. The amadeOS desktop and its applications do not report how you use them. The web browser, Firefox ESR, comes with a configuration that switches its telemetry, studies, Pocket and sponsored content off, makes Ecosia the default search engine and installs the ad blocker uBlock Origin.

Passwords are kept in your keyring. The Mailer stores the passwords of your mail accounts in the desktop's keyring, not in a plain settings file and not on a server.

You decide about updates. amadeOS gets its updates from Debian, a community project, over ordinary package mirrors. Updates are offered, not forced.

The Front Panel shows when updates are waiting, in red when security updates are among them, and installs them when you ask (chapter 52). Automatic security updates are a setting you can turn on; they are off until you do.

You can read every line. All of amadeOS is free software. Anyone can read its source code, check what it does, change it and pass it on. Chapter 4 explains the licence, the EUPL.

Sovereignty is not isolation

amadeOS does not cut you off from the world. It has a web browser, a mail client that talks to every IMAP and SMTP server, Gmail included, and it reads and writes the formats other systems use. The difference is

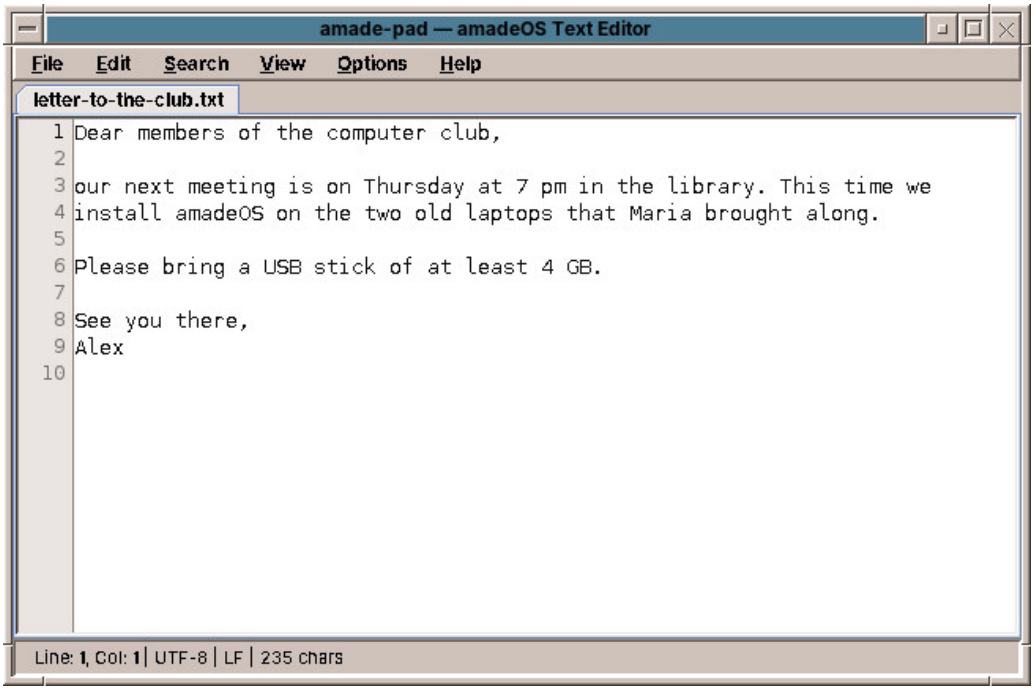
that *you* open each of these doors, one at a time, and you can see which ones are open.

Your data
Throughout this book, boxes like this one say where an application keeps your data and what it sends over the network. If a box is missing somewhere, that is a mistake in the book: please report it.

Why "on your desktop"?

Most talk about digital sovereignty is about servers, clouds and data centres. Those matter. But the place where a person meets the digital world every day is the desktop: the screen, the programs, the files. If the desktop is not under your control, a sovereign data centre behind it helps little.

That is why amadeOS starts there. It does not invent a new kernel or a new way of installing software. It takes proven, free foundations (Linux and Debian) and adds what is missing for sovereignty on the desktop: a complete desktop and its everyday applications, understandable, open and made to last. The next chapter explains why that needs a distribution of its own.



2. Why Another Linux Distribution?

There are hundreds of Linux distributions already. Why one more? This chapter answers in two steps: what a distribution is, and what amadeOS adds that the others do not have.

What a distribution is

Linux itself is only the *kernel*: the part of the system that talks to the hardware. To get a computer you can work with, you need much more: programs to start the system, libraries, a graphical display, a desktop, applications, a way to install and update all of it. A **distribution** puts these thousands of parts together, tests them with each other and delivers them as one system.

Most distributions differ in two things: *how* they deliver software (their package system and release rhythm) and *which desktop* they bring. Many share the same desktops, mostly GNOME and KDE Plasma.

What amadeOS keeps

amadeOS deliberately does **not** reinvent the lower layers. It keeps:

Layer	amadeOS uses	Why
Kernel	Linux, as Debian ships it	Broad hardware support, security fixes, decades of work.
Base system and packages	Debian *trixie* (13), minimal	A community project with a long record of stable releases and security updates; `apt` and `.deb` packages.
Display	Xorg (X11)	Well understood, works with every Java program and with other X11 programs.
Web browser	Firefox ESR from Debian	A free browser with long-term support and security updates.
Sound, network, printing	PipeWire, NetworkManager, CUPS	The standard services of a Linux desktop.

So amadeOS is not a new platform you have to trust from scratch. Underneath, it is Debian. A program packaged for Debian generally installs on amadeOS too, and Debian's security team looks after the base system.

What amadeOS adds

On top of that foundation, amadeOS brings a **desktop of its own**, written from scratch:

- a login screen (the display manager),
- a window manager that draws the window frames,
- the Front Panel, with clock, calendar, workspaces and drawers of applications,
- Settings for colours, text size, backdrops, keyboard, mouse, windows and screen saver,
- tools to look after the system: log viewer, system information, Print Manager, installer,
- and a family of everyday applications: File Manager, Text Editor, WritePad, Mailer, Calendar, Audio Station, Video Player, Image Scan, Fractals and a collection of games.

All of these share one look, one set of colours and one language setting, and they are written in one programming language, Java. That combination is what no other distribution offers, and it cannot simply be

installed on top of another system without its defaults and its integration.

Why not "just a desktop for Debian"?

In a way, amadeOS *is* a desktop for Debian: every part of it is an ordinary Debian package, and Part VII shows how they fit together. But a desktop alone does not make a usable system. Someone has to choose the firmware for common network cards, set up sound and printing, preconfigure the browser for privacy, provide an installer and test that everything works together on a USB stick and on a hard disk. That is the work of a distribution, and amadeOS does it once for everybody.

How amadeOS compares

A rough comparison with a typical desktop distribution based on Debian or Ubuntu with GNOME:

	Typical GNOME distribution	**amadeOS**
Base	Debian or Ubuntu	Debian
Desktop written in	C, JavaScript and more	Java
Look	Modern, changes with major versions	Classic CDE, kept stable on purpose
Applications	GNOME applications, mixed toolkits	amadeOS applications in one style, plus any Linux program
Licence of the desktop	mostly GPL	EUPL-1.2
Maturity	Used by millions	Pre-release (see chapter 6)

The last row matters: amadeOS is young. If you need a system for important work today, chapter 6 tells you honestly what is ready and what is not.

Coming from Windows or macOS
You do not need to know any of the layers in the first table. You will mostly meet the top of the stack: the Front Panel, the windows and the applications, which Parts III to V explain step by step.

3. A Java-Centred Desktop

The most unusual thing about amadeOS is invisible on the screen: the whole desktop is written in **Java**. This chapter explains why, what that means for you, and what is still written in other languages.

One language from the login screen to the editor

On most systems, the parts of the desktop are written in many languages: C for the display server and the window manager, C++, JavaScript, Python or Vala for the panel, the settings and the applications. Each part has its own build system, its own libraries and its own way of drawing on the screen.

In amadeOS, these are all Java programs, built on the same libraries:

Part	Program
Login screen	`amadeos-dm` (display manager and greeter)
Window frames, workspaces, desktop menu	`amadeos-wm` (window manager)
Front Panel	`amadeos-panel-cde`
Settings	`amadeos-settings`
Screen saver and screen lock	`amadeos-session`
System tools	`amadeos-logviewer`, `amadeos-sysinfo`, `amadeos-print`, the installer
Applications and games	`amade-pad`, `amade-mail`, `amade-files`, …, `amade-mines-game`, …

The applications use the shared library *amade-commons* for their colours, their Motif look and their language, so a change in Settings reaches all of them at once.

Why Java?

Understandable. Java is one of the most widely taught programming languages. Many people who have learned to program can read the source code of the amadeOS desktop. A desktop that more people can read is a desktop that more people can check, fix and improve. That is sovereignty in practice, not only in the licence.

Safe by design. Java manages memory itself. A whole class of errors that cause crashes and security holes in C programs (reading past the end of a buffer, using memory after it was freed) cannot happen in Java code.

Made to last. Java programs written twenty years ago still run today. The language changes carefully and keeps old programs working, which fits a desktop that is meant to stay.

Portable. The amadeOS applications and games are built to run on other systems too, on other Linux distributions and on Windows; the games already do. Your favourite tools are not locked to amadeOS either.

Swing. Java comes with a complete toolkit for windows, menus and buttons, *Swing*. amadeOS gives it the look of the Common Desktop Environment and uses it everywhere.

What is not written in Java

amadeOS does not claim that *everything* must be Java. Some parts are better left to proven native software:

- the **Linux kernel** and the drivers for your hardware,
- the **C library** and the basic tools of the system,

- the **Xorg** display server, which draws on the screen and reads keyboard and mouse,
- the **boot loader** GRUB, systemd, NetworkManager, PipeWire and CUPS,
- the **Java runtime** itself (OpenJDK 25 from Debian),
- **Firefox**, the web browser.

The Java programs talk to these parts through their documented interfaces: the window manager speaks the X11 protocol to Xorg, the Front Panel asks NetworkManager about the network, the Print Manager talks to CUPS.

Under the hood
The window manager calls the X11 library `libX11` directly from Java, through *JNA* (Java Native Access), and follows the window-manager conventions ICCCM and EWMH. That is why ordinary X11 programs that are not written in Java, such as `xterm` or Firefox, get the same frames and work with the workspaces. Chapter 61 explains this in detail.

What this means for you

As a user, you notice the Java desktop mostly through its consistency: every window, menu and dialog of amadeOS looks and behaves the same way, follows the same colours and speaks the same language. Programs from other sources (Firefox, LibreOffice, GIMP and so on) work too, with their own look inside the amadeOS window frames.

Coming from Windows or macOS
You do not need Java knowledge to use amadeOS, any more than you need to know C to use Windows. The Java runtime is part of the system and is updated with it.

4. Free, Open and European

amadeOS is free software, it is built on open standards, and it is licensed under a European licence. This chapter explains what each of these means and why it matters to you.

Free software

Free here means freedom, not only price. Free software gives everyone four freedoms:

1. to **run** the program for any purpose, 2. to **study** how it works, because its source code is available, 3. to **share** copies with others, 4. to **improve** it and share the improvements.

All of amadeOS is free software, and so is the Debian system underneath it. No licence fee, no activation, no expiry date, and nobody who can take the software away from you.

The EUPL: a European licence

The amadeOS desktop and its applications are published under the **European Union Public Licence, version 1.2** (EUPL-1.2). It was written and is published by the European Commission.

What makes the EUPL special:

- **It is valid in all official languages of the EU.** There are 23 language versions, and all of them have the same legal value. You can read the licence of amadeOS in your own language, not only in American legal English.
- **It is written for European law.** It was drafted to fit the copyright and contract law of the EU member states.
- **It is a copyleft licence.** Whoever distributes a changed version of amadeOS must publish the changed source code under the EUPL too. Improvements stay free.
- **It is compatible.** The EUPL names a list of other licences (among them the GPL, the LGPL and the MPL) under which code may be combined with it, so amadeOS can work together with the rest of the free-software world.

The text of the licence, in every language, is at interoperable-europe.ec.europa.eu.

Other licences in amadeOS

amadeOS is an *aggregation*: it contains many packages, and each keeps its own licence. A few examples:

Part	Licence
amadeOS desktop, applications, games, this handbook	EUPL-1.2
Linux kernel	GPL-2.0
Debian base system	mostly GPL, LGPL, BSD and MIT
Firefox ESR	MPL-2.0
NsCDE icon theme (package `nscde-icon-theme`)	GPL-3.0
Original CDE backdrops (package `amadeos-backdrops-cde`)	LGPL-2+

On an installed system, the licence of every package is in `/usr/share/doc/<package>/copyright`.

Open standards instead of lock-in

Free code alone is not enough; your *data* must be free too. amadeOS therefore uses open, documented formats and protocols wherever it can:

For	amadeOS uses
Appointments	iCalendar (`.ics`), readable by every calendar program
Mail	IMAP and SMTP, so any mail provider works
Documents	plain text, and OpenDocument (`.odt`) as WritePad's own format
Pictures	PNG, JPEG, TIFF
Applications in the menus	Freedesktop `.desktop` entries
Windows and workspaces	the X11 conventions ICCCM and EWMH
Software	Debian packages (`.deb`)

If you ever leave amadeOS, your data comes with you, and other programs can read it.

Planned
WritePad, the word processor, opens and saves OpenDocument, Word and RTF files only once its import and export steps are done; today it lays out and prints text on pages.

European

amadeOS is developed in Europe, under a European licence, with the European idea of digital sovereignty in mind: public bodies, companies and citizens should be able to choose their software freely and not depend on a handful of vendors from elsewhere. Its desktop speaks five European languages from the start: English, Deutsch, Français, Español and Italiano.

European does not mean closed. amadeOS is open to anyone in the world who wants to use it, study it or help.

5. A Desktop That Stays

Many desktops are redesigned every few years: buttons move, menus disappear, familiar ways of working stop working. amadeOS takes the opposite path. Its desktop follows a design that has proven itself for three decades, and it is meant to stay that way.

Where the look comes from

The **Common Desktop Environment** (CDE) was created in the early 1990s by HP, IBM, Sun and Novell, who agreed on one common desktop for their Unix workstations. For about ten years it was the standard desktop of professional Unix systems such as Solaris, HP-UX and AIX, used in engineering offices, laboratories, banks and universities. In 2012 its source code was released as free software.

CDE was designed for people who work at a computer all day. It is calm, clear and predictable:

- **The Front Panel** at the bottom of the screen holds a clock, a calendar, the most-used programs and the workspace switch. Small arrows above the buttons open *drawers* with more programs.
- **Four workspaces**, *One* to *Four*, each in its own colour, keep the windows of different tasks apart.
- **Windows** have solid frames with three-dimensional edges in the Motif style, a title bar and buttons to minimise and maximise.
- **The Style Manager** changes colours, fonts, backdrop, keyboard, mouse, windows and screen saver in one place.
- **Palettes** colour the whole desktop at once, with matching light and shadow tones.

What amadeOS keeps, and what it changes

amadeOS reproduces this desktop faithfully: the layout of the Front Panel and its drawers, the workspace switch, the window frames, the controls of the Style Manager (called Settings in amadeOS), the palettes and the original CDE backdrops. If you have used CDE, you will feel at home at once.

It is not a museum piece, though. Under the familiar surface amadeOS is a new program, written in Java, and adds what a desktop needs today:

	CDE in the 1990s	amadeOS
Written in	C with the Motif toolkit	Java with Swing
Drawers	Configured by hand	Filled automatically from the installed applications
Mail	Mostly local mailboxes	IMAP with folders, several accounts, safe display of HTML mail
Calendar	Its own server	A plain iCalendar file
Network	Not part of the desktop	Network status in the Front Panel
Fonts	Bitmap fonts	Smooth, scalable fonts

CDE heritage
Boxes like this one appear wherever an amadeOS part follows a CDE program: the login screen follows `dtlogin`, the window manager `dtwm`, Settings `dtstyle`, the Mailer `dtmail`, the Calendar `dtcm`, the Print Manager `dtprintinfo`, the screen saver `dtscreen`. Appendix C lists them all.

Why timeless?

What you learn stays valid. The way you work with amadeOS today is the way you will work with it in ten years. Time spent learning the system is not lost with the next update.

Less distraction. There are no animations, no transparent windows, no blur and no notifications competing for attention. The desktop gets out of the way of your work. amadeOS leaves out such effects on purpose; they are not missing features.

It runs on modest hardware. A desktop without heavy effects needs no expensive graphics card, so older computers stay useful for longer.

Clarity. Every element has a clear edge, a clear state and a clear purpose. Pressed buttons look pressed, the active window is clearly marked, and the colours of the palette tell the workspaces apart.

A desktop that stays does not mean a desktop that stands still. amadeOS gets new applications, new settings and fixes all the time. What it avoids is change for its own sake.